Chapter 2
Boundary-Value Problems in Electrostatics: I

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These notes follow the notation and viewpoint of Jackson on pp. 57–84. I keep the same symbols used in the text whenever possible: scalar potential Φ, charge density ρ, surface-charge density σ, Green function $G(\mathbf{x}, \mathbf{x}')$, and SI units with ε_0. The emphasis here is on what each construction is really doing physically, not only on the final formulas.

Prepared from Chapter 2 of *Classical Electrodynamics*.

Contents

1 Method of Images **2**

2 Point Charge in the Presence of a Grounded Conducting Sphere **3**

3 Point Charge in the Presence of a Charged, Insulated Conducting Sphere **4**

4 Point Charge Near a Conducting Sphere at Fixed Potential **4**

5 Conducting Sphere in a Uniform Electric Field by Method of Images **5**

6 Green Function for the Sphere; General Solution for the Potential **6**

7 Conducting Sphere with Hemispheres at Different Potentials **6**

8 Orthogonal Functions and Expansions **7**

9 Separation of Variables; Laplace Equation in Rectangular Coordinates **8**

10 A Two-Dimensional Potential Problem; Summation of a Fourier Series **8**

11 Fields and Charge Densities in Two-Dimensional Corners and Along Edges **9**

12 Introduction to Finite Element Analysis for Electrostatics **10**

Guiding idea of the chapter

Chapter 1 gave the formal solution of Poisson and Laplace problems through Green functions. Chapter 2 is where that formalism starts becoming useful in practice. Jackson focuses on three strategies:

1. replacing conductors by fictitious charges through the method of images,
2. representing solutions in orthogonal bases adapted to the geometry,
3. introducing numerical ideas through finite elements.

The common theme is simple: the field equation is local, but the boundary conditions encode the real geometry of the problem. Every method in this chapter is really a way of making the boundary conditions manageable.

1. Method of Images

The method of images is one of the cleanest examples of how uniqueness theorems turn a clever guess into an exact solution. The electrostatic problem usually begins with a point charge in the presence of one or more conducting surfaces, grounded or held at fixed potential. Instead of solving Laplace's equation with those surfaces present explicitly, one enlarges the region and replaces the conductors by fictitious charges placed outside the physical region of interest.

The logic is as follows. Inside the physical region, away from the real charges, the potential must satisfy Laplace's equation. If I can invent an auxiliary configuration of charges outside that region whose potential satisfies the same differential equation inside and also reproduces the required boundary values on the conductor, then by uniqueness the auxiliary potential is the physical one.

The method works only when the geometry is favorable. A plane, a sphere, and a few other highly symmetric boundaries allow the image charges to be guessed in finite number. In more complicated geometries the Green-function language remains exact, but the images are no longer easy to identify or may require infinite sets.

The simplest example is a charge q in front of a grounded infinite plane. The physical problem is equivalent to replacing the plane by a mirror charge $-q$ at the reflected point. The potential in the allowed half-space is then the superposition of the real charge and the image charge. The conductor disappears from the calculation, but its boundary condition remains built into the construction.

What matters conceptually is that image charges are not physically present. They are bookkeeping devices that manufacture the correct harmonic correction to the potential.

2. Point Charge in the Presence of a Grounded Conducting Sphere

Jackson's first major spherical example is a point charge q located at distance y from the center of a grounded conducting sphere of radius a. The goal is to find the potential outside the sphere subject to

$$\Phi(|\mathbf{x}| = a) = 0.$$

By symmetry, if a single image charge is enough, it must lie on the same ray from the center as the real charge. One therefore tries

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{|\mathbf{x}-\mathbf{y}|} + \frac{q'}{|\mathbf{x}-\mathbf{y}'|}\right).$$

Imposing the vanishing of the potential on the sphere determines the image data uniquely:

$$q' = -\frac{a}{y}q, \qquad y' = \frac{a^2}{y}.$$

So the image charge lies inside the sphere, with opposite sign and reduced magnitude.

This result is elegant because the geometry of the sphere turns the boundary condition into an inversion relation. The image point is not merely a reflected point as in the plane case; it is the inverse point under the mapping $yy' = a^2$. This inversion structure reappears later when Jackson writes the Green function for the sphere.

Once the potential is known, the induced surface-charge density follows from the normal derivative at the conductor surface. Jackson obtains

$$\sigma = -\varepsilon_0 \left.\frac{\partial\Phi}{\partial x}\right|_{x=a} = -\frac{q}{4\pi a^2}\left(\frac{a}{y}\right)\frac{1 - a^2/y^2}{(1 + a^2/y^2 - 2(a/y)\cos\gamma)^{3/2}},$$

where γ is the angle between the observation point on the sphere and the radius toward the charge.

The qualitative behavior is very clear. The surface charge is strongly concentrated on the near side of the sphere. As the external charge approaches the surface, the induced density becomes more and more peaked. The total induced charge is equal to the magnitude of the image charge, which is exactly what Gauss's law demands.

The force on the real charge is easiest to obtain by treating the image charge as if it were physical. One gets the attractive force

$$|\mathbf{F}| = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{a^2}\left(\frac{a}{y}\right)^3\left(1 - \frac{a^2}{y^2}\right)^{-2}.$$

Far away the force falls as y^{-3}, the signature of an induced dipole interaction. Very near the sphere it grows much more sharply because the local induced charge distribution becomes strongly nonuniform.

An important remark in Jackson is that the same formulas can be translated to the case $y \leq a$,

when the charge lies inside the sphere, provided one reinterprets the outward normal correctly. The mathematics is the same; only the physical side of the boundary changes.

3. Point Charge in the Presence of a Charged, Insulated Conducting Sphere

Now the sphere is no longer grounded. Instead it carries a specified total charge Q. Jackson builds this problem by superposition. Start with the grounded-sphere solution. That already accounts for the induced redistribution needed to enforce an equipotential surface. Then add enough charge to the sphere to bring its total charge from q' to the desired value Q.

Because the added charge lives on a conducting sphere, it distributes uniformly. Outside the sphere, that extra piece of the potential is the same as the potential of a point charge at the center. Thus the total potential becomes

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{|\mathbf{x}-\mathbf{y}|} - \frac{aq}{y\left|\mathbf{x}-\frac{a^2}{y^2}\mathbf{y}\right|} + \frac{Q+\frac{a}{y}q}{|\mathbf{x}|}\right].$$

The force on the point charge is then

$$\mathbf{F} = \frac{1}{4\pi\varepsilon_0}\frac{q}{y^2}\left[Q - \frac{qa^3(2y^2-a^2)}{y(y^2-a^2)^2}\right]\frac{\mathbf{y}}{y}.$$

This example is physically rich. Even if the sphere has the same sign as the point charge, the force can still be attractive when the point charge is close enough, because the induced surface polarization dominates the direct Coulomb repulsion from the net charge. Only far away does the force reduce to the naive Coulomb form for two small charged bodies. Near the surface, the conductor behaves less like a point object and more like a deformable charge distribution.

This is one of the nicest illustrations of why conductors can hold excess charge despite mutual repulsion: removing a bit of surface charge from the conductor costs work because the image attraction tends to pull it back.

4. Point Charge Near a Conducting Sphere at Fixed Potential

Instead of specifying the total charge Q, one may prescribe the potential V of the sphere. The construction is almost the same as in the previous section. The grounded-sphere contribution is retained, but now the central correction is chosen so that the potential on $|\mathbf{x}| = a$ equals V.

The result is

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{|\mathbf{x}-\mathbf{y}|} - \frac{aq}{y\left|\mathbf{x}-\frac{a^2}{y^2}\mathbf{y}\right|}\right] +$$

racVa|x|.

The force on the external charge is

$$\mathbf{F} = \frac{q}{y^2}\left[Va - \frac{1}{4\pi\varepsilon_0}\frac{qay^3}{(y^2-a^2)^2}\right]\frac{\mathbf{y}}{y}.$$

The structure is the same as before, but conceptually this problem is cleaner if one thinks in terms of boundary conditions rather than conserved total charge. The conductor is connected to a reservoir that can add or remove charge as needed so long as the surface remains at potential V.

For corresponding values of V and Q, the force law resembles the previous one closely, but the approach to the far-field limit is somewhat different. Again there can be an unstable equilibrium point where attraction and repulsion balance.

5. Conducting Sphere in a Uniform Electric Field by Method of Images

Jackson next treats a conducting sphere in a uniform applied field E_0. A uniform field may be imagined as the field created near the origin by two distant charges $\pm Q$ at $z = \mp R$, with Q/R^2 fixed as $R \to \infty$. After inserting a conducting sphere of radius a at the origin, one adds the corresponding images of those distant charges and finally takes the large-R limit.

The result is the compact potential

$$\Phi(r,\theta) = -E_0\left(r - \frac{a^3}{r^2}\right)\cos\theta.$$

The first term is simply the background uniform field. The second is the potential of an induced dipole. This is why a neutral conductor in a uniform field behaves, from far away, as if it had acquired dipole moment

$$\mathbf{p} = 4\pi\varepsilon_0 a^3 E_0\,\hat{\mathbf{z}}.$$

The induced surface-charge density is obtained from the radial derivative at $r = a$:

$$\sigma = 3\varepsilon_0 E_0\cos\theta.$$

It integrates to zero, as it should for a neutral isolated sphere. In fact there is no difference between grounded and insulated here because the total induced charge vanishes automatically.

This is one of the classic polarizability results. The sphere responds to a uniform field by creating precisely the dipole correction needed to enforce that its surface is an equipotential.

6. Green Function for the Sphere; General Solution for the Potential

The image solution for the sphere can be reinterpreted as a Green function. For Dirichlet boundary conditions on a sphere of radius a, the Green function is the potential of a unit source at $\mathbf{x}'$ together with its image chosen so that $G = 0$ on the sphere. Jackson writes it as

$$G(\mathbf{x}, \mathbf{x}') = \frac{1}{|\mathbf{x} - \mathbf{x}'|} - \frac{a}{x' \left|\mathbf{x} - \frac{a^2}{x'^2}\mathbf{x}'\right|}.$$

In spherical form,

$$G(\mathbf{x}, \mathbf{x}') = \frac{1}{(x^2 + x'^2 - 2xx' \cos\gamma)^{1/2}} -$$

rac1$\left(\frac{x^2 x'^2}{a^2} + a^2 - 2xx' \cos\gamma\right)^{1/2}$, where γ is the angle between $\mathbf{x}$ and $\mathbf{x}'$.

This expression is symmetric in $\mathbf{x}$ and $\mathbf{x}'$, as a Dirichlet Green function should be. More importantly, it converts the image construction from a special trick into a general integral formula. The normal derivative on the boundary enters directly into the representation of the potential outside the sphere:

$$\Phi(\mathbf{x}) = \frac{1}{4\pi} \int \Phi(a, \theta', \phi') \frac{a(x^2 - a^2)}{(x^2 + a^2 - 2ax \cos\gamma)^{3/2}} \, d\Omega'.$$

So the earlier image problem is now seen as one concrete realization of the formal Green-function method of Chapter 1.

7. Conducting Sphere with Hemispheres at Different Potentials

Jackson then applies the spherical Green function to a sphere split into two hemispheres held at potentials $+V$ and $-V$. This is a particularly instructive example because it is simple enough to set up exactly, yet complicated enough that a closed form for the general point is not immediate.

Using the boundary integral formula, the potential outside the sphere is written as an angular integral over the upper hemisphere minus the lower hemisphere. On the positive z-axis, where the geometry simplifies, the integral can be done explicitly and gives

$$\Phi(z) = V \left[1 - \frac{z^2 - a^2}{z\sqrt{z^2 + a^2}}\right].$$

Far from the sphere the field behaves like a dipole field, as symmetry demands. Since the net boundary potential is antisymmetric under $z \to -z$, only odd angular structures survive.

Jackson then expands the full solution in powers of the dimensionless parameter

$$\alpha = \frac{ax}{a^2 + x^2},$$

which produces only odd powers of $\cos\theta$. The leading terms show explicitly that the potential is

dominated by $P_1(\cos\theta)$ and then receives corrections from $P_3(\cos\theta)$, $P_5(\cos\theta)$, and so on. This is a nice preview of why Legendre polynomials are the natural language for separated solutions in spherical geometry.

8. Orthogonal Functions and Expansions

Jackson pauses here to review orthogonal-function expansions because boundary-value methods will depend on them repeatedly. If $U_n(\xi)$ is an orthonormal set on (a, b), then

$$\int_a^b U_n^*(\xi) U_m(\xi)\, d\xi = \delta_{nm}.$$

Any square-integrable function can then be expanded as

$$f(\xi) = \sum_{n=1}^{\infty} a_n U_n(\xi), \qquad a_n = \int_a^b U_n^*(\xi) f(\xi)\, d\xi.$$

The expansion coefficients minimize the mean-square error. That is why orthogonal bases are not merely convenient; they are optimal in a precise metric sense.

Completeness introduces the closure relation

$$\sum_{n=1}^{\infty} U_n^*(\xi') U_n(\xi) = \delta(\xi' - \xi).$$

This is the continuous analogue of a basis decomposition in linear algebra.

Jackson then reviews Fourier series,

$$f(x) = \frac{1}{2} A_0 + \sum_{m=1}^{\infty} \left[A_m \cos\left(\frac{2\pi m x}{a}\right) + B_m \sin\left(\frac{2\pi m x}{a}\right) \right],$$

and the Fourier integral,

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} A(k) e^{ikx}\, dk, \qquad A(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ikx} f(x)\, dx.$$

The message is that the choice of basis is controlled by symmetry and domain. Once the geometry suggests the right eigenfunctions, the boundary conditions determine the coefficients.

9. Separation of Variables; Laplace Equation in Rectangular Coordinates

For Laplace's equation in Cartesian coordinates,

$$\frac{\partial^2\Phi}{\partial x^2}+\frac{\partial^2\Phi}{\partial y^2}+\frac{\partial^2\Phi}{\partial z^2}=0,$$

Jackson assumes a product form

$$\Phi(x,y,z)=X(x)Y(y)Z(z).$$

After division by XYZ, each coordinate dependence must separately equal a constant. One writes

$$\frac{1}{X}\frac{\mathrm{d}^2X}{\mathrm{d}x^2}=-\alpha^2,\qquad \frac{1}{Y}\frac{\mathrm{d}^2Y}{\mathrm{d}y^2}=-\beta^2,\qquad \frac{1}{Z}\frac{\mathrm{d}^2Z}{\mathrm{d}z^2}=\gamma^2,\qquad \alpha^2+\beta^2=\gamma^2.$$

This produces sines, cosines, exponentials, or hyperbolic functions depending on which boundaries are to be satisfied.

Jackson's representative problem is a rectangular box with five grounded faces and the sixth face at prescribed potential $V(x,y)$. The solution is expanded as

$$\Phi(x,y,z)=\sum_{n,m=1}^{\infty}A_{nm}\sin\left(\frac{n\pi x}{a}\right)\sin\left(\frac{m\pi y}{b}\right)\sinh(\gamma_{nm}z),$$

with

$$\gamma_{nm}=\pi\sqrt{\frac{n^2}{a^2}+\frac{m^2}{b^2}}.$$

Matching the boundary value at $z=c$ yields the coefficients

$$A_{nm}=\frac{4}{ab\sinh(\gamma_{nm}c)}\int_0^a\int_0^b V(x,y)\sin\left(\frac{n\pi x}{a}\right)\sin\left(\frac{m\pi y}{b}\right)dx\,dy.$$

This is the canonical separated solution: the geometry dictates the eigenfunctions, and the nonhomogeneous boundary data appear only in the expansion coefficients.

10. A Two-Dimensional Potential Problem; Summation of a Fourier Series

Jackson next turns to the two-dimensional strip-like region

$$0\le x\le a,\qquad y\ge 0,$$

with $\Phi = 0$ on $x = 0$ and $x = a$, $\Phi = V$ on $y = 0$, and $\Phi \to 0$ as $y \to \infty$. Separation of variables gives

$$\Phi(x, y) = \sum_{n=1}^{\infty} A_n e^{-n\pi y/a} \sin\left(\frac{n\pi x}{a}\right),$$

and matching the constant boundary value at $y = 0$ leaves only odd terms,

$$A_n = \frac{4V}{\pi n}, \qquad n \text{ odd}.$$

Hence

$$\Phi(x, y) = \frac{4V}{\pi} \sum_{n\,\text{odd}} \frac{1}{n} e^{-n\pi y/a} \sin\left(\frac{n\pi x}{a}\right).$$

This is already useful, but Jackson goes further and sums the series exactly by introducing

$$Z = e^{(i\pi/a)(x+iy)}.$$

Then the odd series becomes a logarithm, and the potential can be written in closed form as

$$\Phi(x, y) = \frac{2V}{\pi} \tan^{-1}\left(\frac{\sin(\pi x/a)}{\sinh(\pi y/a)}\right).$$

This is a beautiful example of how separation of variables, Fourier analysis, and complex-variable structure are all different faces of the same harmonic problem.

11. Fields and Charge Densities in Two-Dimensional Corners and Along Edges

One of the most important local lessons of electrostatics is that sharp corners and edges generate singular fields. Jackson studies the idealized geometry of two conducting planes meeting at opening angle β, both held at potential V. In polar coordinates, Laplace's equation is

$$\frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial \Phi}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 \Phi}{\partial \phi^2} = 0.$$

A separated form $\Phi(\rho, \phi) = R(\rho)\Psi(\phi)$ yields radial powers and angular sines/cosines. Enforcing the constant potential on the two conducting faces gives the near-corner expansion

$$\Phi(\rho, \phi) \simeq V + a_1 \rho^{\pi/\beta} \sin\left(\frac{\pi\phi}{\beta}\right).$$

The electric-field components therefore behave as

$$E_\rho \sim \rho^{(\pi/\beta)-1}, \qquad E_\phi \sim \rho^{(\pi/\beta)-1},$$

and the surface-charge density on the conducting faces has the same power-law behavior,

$$\sigma(\rho) \sim \rho^{(\pi/\beta)-1}.$$

This single exponent explains a lot of physics. If $\beta < \pi$, the exponent is positive and the field actually dies away at the corner. For a flat surface, $\beta = \pi$, the field remains finite. If $\beta > \pi$, corresponding to an edge rather than an interior corner, the exponent is negative and the field diverges as the edge is approached. For the edge of a thin conducting sheet, $\beta = 2\pi$, so

$$\sigma(\rho) \sim \rho^{-1/2}.$$

The singularity is still integrable, so the total charge near the edge remains finite, but the local field becomes very large.

This is not a mathematical curiosity. Jackson explicitly connects it with the practical effectiveness of pointed conductors and lightning rods. Sharp edges intensify the field enough to trigger breakdown first in their neighborhood.

12. Introduction to Finite Element Analysis for Electrostatics

The final section of the chapter introduces finite element analysis (FEA) through Galerkin's method. Jackson starts from the two-dimensional Poisson equation

$$\nabla^2 \psi = -g$$

in a region R with Dirichlet boundary conditions. Multiplying by a test function ϕ that vanishes on the boundary and integrating by parts gives the weak form

$$\int_R [\nabla\phi \cdot \nabla\psi - g\phi]\, dx\, dy = 0.$$

The key move is to approximate the unknown field by a finite expansion in localized basis functions.

Jackson first discusses a square-lattice choice,

$$\phi_{ij}(x, y) = \left(1 - \frac{|x - x_i|}{h}\right)\left(1 - \frac{|y - y_j|}{h}\right),$$

for $|x - x_i| \leq h$, $|y - y_j| \leq h$, and zero otherwise. The approximate field is

$$\psi(x, y) \approx \sum_{k,l} \Psi_{kl}\phi_{kl}(x, y).$$

Choosing the test function to be one basis function at a time leads to a sparse linear system. Only nearby nodes couple to each other, which is the basic numerical advantage of localized bases.

Jackson then moves to the more flexible triangular element. On a triangle with vertices (x_1, y_1),

(x_2, y_2), (x_3, y_3), the field is approximated linearly:

$$\psi_e(x, y) = A + Bx + Cy.$$

It is more systematic to express this through shape functions $N_i^{(e)}(x, y)$, one for each vertex, such that $N_i^{(e)} = 1$ at vertex i and 0 at the other two. Then within one element,

$$\psi(x, y) \approx \sum_{j=1}^{3} \Psi_j^{(e)} N_j^{(e)}(x, y).$$

The element matrix takes the form

$$k_{ij}^{(e)} = S_e(b_i b_j + c_i c_j),$$

where S_e is the triangle area and the coefficients b_i, c_i come from the linear shape functions.

For each element, the Galerkin equations become

$$\sum_{j=1}^{3} k_{ij}^{(e)} \Psi_j^{(e)} = \frac{S_e}{3} g_e, \qquad i = 1, 2, 3,$$

with g_e approximated at the centroid. Assembling all elements gives the global sparse matrix equation

$$\mathbf{K}\boldsymbol{\Psi} = \mathbf{G}.$$

The diagonal entries of $\mathbf{K}$ collect contributions from all triangles sharing a node, and the off-diagonal entries collect couplings between nodes connected by element edges.

This section is brief, but it already contains the central philosophy of FEA: one does not solve the differential equation pointwise. One solves a weak integral statement in a finite-dimensional space built from local shape functions. The geometry can then be refined where needed, which is precisely why finite elements are so much more adaptable than a uniform square grid.

Jackson ends by remarking that the three-dimensional generalization is immediate in spirit: one replaces triangular elements by tetrahedra and uses four shape functions instead of three.

Closing synthesis

Chapter 2 is where electrostatics stops looking like a collection of isolated textbook tricks and starts becoming a real theory of boundary-value problems. The method of images shows how symmetry can collapse a boundary into a few fictitious charges. Orthogonal expansions show how geometry selects a natural basis in which the boundary data can be decomposed. Separation of variables reveals that the admissible functional forms are not arbitrary but are eigenfunctions of the coordinate-separated operators. The corner analysis shows that local geometry controls singular structure. And the finite-element discussion shows how all of this survives numerically once exact analytic forms are no longer available.

To me, the most useful lesson in this chapter is that the field equation is usually the easy part. The hard part is always the boundary.